



MECH MAINT- OFF SITE Department  
SIMHADRI

**Note No :1**



**Ref:**SMTTP/MECH MAINT-/2023-24/LMIBFV23/751961

**Date:**

20/12/2023(15:02:52)

**Subject:**LMI Revision -6 Requirements for the Operation and Maintenance of Large Butterfly Valves in Cooling Water Systems at NTPC Simhadri

LMI on Requirements for the Operation and Maintenance of Large Butterfly Valves in Cooling Water Systems at NTPC Simhadri is revised as required & same is requested to review by concerned department .

**LMI Rev-6 Requirements for the Operation and Maintenance of Large Butterfly Valves in Cooling Water Systems at NTPC Simhadri , with reference : OGN-OPS-MECH-008\_Rev4\_Requirements for the OnM of Large Butterfly Valves in Cooling Water Systems is attached .**

Submitted for review by concerned HOD's & Required approval for implementation by competent authority .

**Submitted by:**

डी.श्री रमा सास्त्री - उप महाप्रबंधक

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**On:**20/12/2023(15:02:52) **Ref:**SMTTP/MECH MAINT-/2023-24/LMIBFV23/751961

**Note No:2**

In line with COS-ISO-00-OGN/OPS/MECH/008 Rev. No.: 3 March 2023, revised LMI (LMI/OGN/OPS/MECH/008 Rev- 5) for Operation and Maintenance of Large Butterfly Valves in Cooling Water Systems at NTPC Simhadri vide Pradip Ref No-SMTPP/MECH MAINT-/2023-24/LMILBFV5/211749 dated 14/06/2023 was approved by HOP, Simhadri.

Again on revision of OGN vide Ref No : COS-ISO-00-OGN/OPS/MECH/008 Rev. No.: 4 Oct. 2023, revision-6 of the LMI is being forwarded for persusal of all concerned and subsequent approval of HOP, Simhadri, please.

**Submitted by:**

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**On:**20/12/2023(17:18:39) **Ref:**SMTPP/MECH MAINT-/2023-24/LMIBFV23/751961

**Note No:3**

**Submitted by:**

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**On:**21/12/2023(15:53:34) **Ref:**SMTPP/MECH MAINT-/2023-24/LMIBFV23/751961

**Note No:4**

**Submitted by:**

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**On:**26/12/2023(10:31:18) **Ref:**SMTPP/MECH MAINT-/2023-24/LMIBFV23/751961

**Note No:5**

Please review

**Submitted by:**

नवीन किशोर - अपर महाप्रबंधक

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**On:**29/12/2023(11:36:43) **Ref:**SMTPP/MECH MAINT-/2023-24/LMIBFV23/751961

**Note No:6**

LMI reviewed and no changes proposed.

Put up for further approval.

**Submitted by:**

रश्मि रंजन सतपथी - वरिष्ठ प्रबंधक

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**On:**01/01/2024(19:21:46) **Ref:**SMTTPP/MECH MAINT-/2023-24/LMIBFV23/751961

**Note No:7**

LMI reviewed and no changes proposed. Forwarded for further approval.

**Submitted by:**

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**On:**01/01/2024(20:30:12) **Ref:**SMTTPP/MECH MAINT-/2023-24/LMIBFV23/751961

**Note No:8**

Kindly review the LMI

**Submitted by:**

जे आर वर्गीस - अपर महाप्रबंधक

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**On:**03/01/2024(09:29:32) **Ref:**SMTTPP/MECH MAINT-/2023-24/LMIBFV23/751961

**Note No:9**

Kindly review the LMI

**Submitted by:**

टी वाई रेड्डी - उप महाप्रबंधक

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**On:**05/01/2024(09:38:42) **Ref:**SMTTPP/MECH MAINT-/2023-24/LMIBFV23/751961

**Note No:10**

LMI reviewed and following changes proposed please:

As per COS OGN "COS-ISO-00-OGN/OPS/MECH/008 Rev. No.: 04, Oct. 2023" healthiness of PRV's, ball valves, drain valves etc. is to be ensured by MM, accordingly responsibility to be changed in LMI from C&I to MM in Annexure- 1 Page No:9.

**Submitted by:**

चन्दन कुमार निराला - वरिष्ठ प्रबंधक

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**On:**09/01/2024(10:08:55) **Ref:**SMTTP/MECH MAINT-/2023-24/LMIBFV23/751961

**Note No:11**

Submitted for changes proposed by Chandan Kumar Nirala

**Submitted by:**

टी वाई रेड्डी - उप महाप्रबंधक

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**On:**09/01/2024(10:14:04) **Ref:**SMTTP/MECH MAINT-/2023-24/LMIBFV23/751961

**Note No:12**

**Submitted by:**

जे आर वर्गीस - अपर महाप्रबंधक

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**On:**09/01/2024(10:52:05) **Ref:**SMTTP/MECH MAINT-/2023-24/LMIBFV23/751961

**Note No:13**

**Submitted by:**

धर्मेन्द्र कुमार तोमर - अपर महाप्रबंधक

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**On:**10/01/2024(09:40:25) **Ref:**SMTTP/MECH MAINT-/2023-24/LMIBFV23/751961

**Note No:14**

LMI reviewed.

Forwarded for further

**Submitted by:**

कृष्णा बीसेट्टी - उप महाप्रबंधक

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**On:**10/01/2024(10:20:19) **Ref:**SMTTP/MECH MAINT-/2023-24/LMIBFV23/751961

**Note No:15**

LMI is reviewed and found in line with OGN.

**Submitted by:**

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**On:**13/01/2024(10:01:05) **Ref:**SMTTPP/MECH MAINT-/2023-24/LMIBFV23/751961

**Note No:16**

**Submitted by:**

टी.आर. राजगोपालन् - महाप्रबंधक

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**On:**15/01/2024(10:02:41) **Ref:**SMTTPP/MECH MAINT-/2023-24/LMIBFV23/751961

**Note No:17**

May please be accorded approval.

**Submitted by:**

मैथ्यू ई कोवूर - महाप्रबंधक

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**On:**23/01/2024(15:40:13) **Ref:**SMTTPP/MECH MAINT-/2023-24/LMIBFV23/751961



**Note No:18**

**Approved & Submitted by:**

संजय कुमार सिन्हा - महाप्रबंधक

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**On:**23/01/2024(18:46:25) **Ref:**SMTPP/MECH MAINT-/2023-24/LMIBFV23/751961

NTPC

SIMHADRI

## **LOCATION MANAGEMENT INSTRUCTION**

LMI/OGN/OPS/MECH/008

REVISION NO. 6.00; DATE: DEC - 2023

# ***REQUIREMENTS FOR THE OPERATION AND MAINTENANCE OF LARGE BUTTERFLY VALVES IN COOLING WATER SYSTEMS AT NTPC SIMHADRI***

APPROVED FOR

*Implementation-----signed-----*

**HOP- SIMHADRI**

**Date-----**

[Type text]

**LMI APPROVAL DOCUMENTS**

|                                                                    |                                                                                                                                       |
|--------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>1.0 TITLE OF LMI</b>                                            | <b>Requirements for the Operation and Maintenance of Large Butterfly Valves in Cooling Water Systems at NTPC Simhadri</b>             |
| <b>2.a) DOCUMENT CODE</b><br><b>b) REVISION/ISSUE NO.&amp;DATE</b> | <b>LMI/OGN/OPS/MECH/008/</b><br><b>Rev: 6.00; Dec '2023</b>                                                                           |
| <b>3.ORIGINAL/REF.DOCUMENT</b>                                     | Ref No.: EOC/OS/2022-23/COSTCOGN/899194 (dt: 21.03.2023)<br><b>COS-ISO-00-OGN/OPS/MECH/008;</b><br><b>Rev No: 5, Date: June '2023</b> |
| <b>4.PRERPREPARED BY</b>                                           | <b>NAME: D.SRI RAMA SASTRY, DGM (MM-OFS)</b><br><b>SIGNATURE-----</b>                                                                 |
| <b>5. CHECKED BY</b>                                               | <b>NAME: ASIS KUMAR MUKHOPADHYAY</b><br><b>AGM(MM-OFS&amp; CIVIL)</b><br><b>SIGNATURE-----</b>                                        |
| <b>6.a)RECHECKED BY</b>                                            | <b>NAME: SHAMBHU KUMAR SUMAN, AGM(TMD)</b><br><b>SIGNATURE-----</b>                                                                   |
| <b>6.b) RECHECKED BY</b>                                           | <b>NAME: Jayamol Rachel Varughese AGM ( CnI)</b><br><b>SIGNATURE-----</b>                                                             |
| <b>6.c) RECHECKED BY</b>                                           | <b>NAME: A.K.TOMAR , AGM(EMD)</b><br><b>SIGNATURE-----</b>                                                                            |
| <b>6.d ) RECHECKED BY</b>                                          | <b>NAME: T RAJA GOPALAN, GM ( O)</b><br><b>SIGNATURE-----</b>                                                                         |
| <b>6.e) REVIEWED BY</b>                                            | <b>NAME: MATHEW EIPE KOVOOR , GM(MNTCE)</b><br><b>SIGNATURE-----</b>                                                                  |
| <b>7. APPROVED BY</b>                                              | <b>NAME: DHURJATI PRASAD PATRA</b><br><b>SIGNATURE:-----</b><br><b><i>GM(O&amp;M), SIMHADRI</i></b>                                   |
| <b>8.DATE OF ISSUE</b>                                             |                                                                                                                                       |
| <b>9.INDEX LIST</b>                                                | <b>ENCLOSED.</b>                                                                                                                      |
| <b>10.DISTRIBUTION LIST</b>                                        | <b>ENCLOSED.</b>                                                                                                                      |

|                                                                                                                                                                    |                                                                   |                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|-----------------------|
| <b>NTPC</b>                                                                                                                                                        | <b>LMI</b>                                                        | <b>SIMHADRI</b>       |
| <b>TITLE: Requirements for the Operation and Maintenance of Large Butterfly Valves in Cooling Water Systems at NTPC Simhadri      Rev: 6.00      Date:Dec'2023</b> |                                                                   |                       |
| <b>REF NO. LMI/OGN/OPS/MECH/008/</b>                                                                                                                               |                                                                   | <b>PAGE : 1 OF 10</b> |
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| NTPC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | LMI | SIMHADRI                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------------------------|
| TITLE: Requirements for the Operation and Maintenance of Large Butterfly Valves in Cooling Water Systems at NTPC Simhadri                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |     | Rev: 6.00<br>Date:Dec'2023 |
| REF NO. LMI/OGN/OPS/MECH/008/                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     | PAGE : 2 OF 10             |
| <p><b><u>2.0 INTRODUCTION</u></b></p> <p>The large butterfly valves of 2400 MM dia(CW P/H), 2200MMdia(STG-2 Condenser),2100 MM dia (Stg-1 condenser) are provided in the CW Systems at NTPC Simhadri. These are Hydraulic operated &amp; Motor operated valves for full open and close. These valves are provided with lock pin arrangement for locking in open and close position.</p> <p>There have been reported incidents at other power stations, on failure of lock pin due to play (free movement) between valve disc and locking pin, as a result of wear induced by disc flutter. This free movement permitted large inertia forces to be applied to the pin, which was the dominant factor in its failure. An occurrence with consequent hydraulic shocks resulting in severe structural damage to valve seat and body &amp; also valve getting stuck up during opening &amp; closing observed which lead to generation loss in few NTPC stations.</p> <p>The objective of this LMI is to establish a inspection strategy for Hydraulically &amp; Motor operated large butterfly valves of size 2400 NB ,2200NB, 2100NB diameter in the CW Systems of stage-1 &amp;2 in order to prevent their failure and ensure safety.</p> <p><b><u>3.0 SCOPE</u></b></p> <p>The scope of this LMI is to detail the methods and periodicity that are to be adopted for the inspection of 5 nos of BFV- 2400 mm dia, (STG1 CW P/H), 5 Nos BFV-2100 mm dia, Stg-1 Condenser), 5 nos BFV 2200 mm dia (Stg-2 CW P/H) &amp;16 nos of BFV-2200 mm dia (Stg-2 Condenser). Hydraulically &amp; motor operated Butterfly valves in CW Systems of stage-1 &amp;2</p> <p><b><u>4.0 REFERENCE DOCUMENT:</u></b></p> <p>Operation guidance note No. COS-ISO-00-OGN/OPS/MECH/008 (Rev. No:04, Date :Oct 2023) circulated by operation services- NTPC corporate center.</p> <p><b><u>5.0 SUPERSEDED DOCUMENT</u></b></p> <p>Requirements for the Operation and Maintenance of Large Butterfly Valves in Cooling Water Systems at NTPC Simhadri. Ref no: ISSUE NO.3; Rev. No:5, Date: June -2023</p> <p>Revision Details: Extended the Applicability of LMI &amp; same is adopted for the inspection of butterfly valves of BFV-2100 mm dia ( in Stg-1 Condenser), 5 nos BFV 2200 mm dia (in Stg-2 CW P/H) &amp; 16 nos of BFV-2200 mm (in Stg-2 Condenser) diameter. Hydraulically&amp; motor operated Butterfly valves in CW Systems of stage-1 &amp;2</p> <p><b><u>6.0 SPECIFIC INSTRUCTIONS</u></b></p> <p>To achieve the objective of the LMI the following are to be followed:</p> <ol style="list-style-type: none"> <li>Periodic inspection of valves and valve restraint mechanisms.</li> <li>Valve locking should be done properly to avoid free rotation.</li> <li>Remedial actions to sustain reliable continued operation on discovery of defects.</li> </ol> |     |                            |

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| <b>NTPC</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>LMI</b>       | <b>SIMHADRI</b>      |
| <b>TITLE: Requirements for the Operation and Maintenance of Large Butterfly Valves in Cooling Water Systems at NTPC Simhadri,</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                  |                      |
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| <b>REF NO. LMI/OGN/OPS/MECH/008/</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                  | <b>PAGE :3 OF 10</b> |
| <p><b><u>7.0 METHODOLOGY OF IMPLEMENTATION AND RESPONSIBILITY CENTRES</u></b></p> <p>A) During every Unit overhaul, the freeness between the locking pin and the blade restraint is to be checked. If the free movement is more than 5 degrees, then suitable rectification measure such as changing the pin are to be taken.<br/>Responsibility: Head of MM-OS, Head of MM-TMD</p> <p>B) Any frequent occurrences of free movement between locking pin and blade restraint mechanism should be referred to OEM.<br/>Responsibility: Head of MM-OS, Head of MM-TMD</p> <p>C) The butterfly valves internals should be checked during every unit overhaul for integrity of seals, blade and blade operating mechanisms.<br/>Responsibility: Head of MM-OS, Head of MM-TMD</p> <p>D) Where no excess free movement has been found the valve should be rechecked at a convenient opportunity.<br/>Responsibility: Head of MM-OS, Head of MM-TMD</p> <p>E) Healthiness of check valves, Flow control valves, Solenoid Valves, Valve position feed back, Limit Switches, , Noting the timing of valves.<br/>Responsibility: Head of C&amp;I</p> <p>F) Healthiness of lever engagement for manual operation of actuator valve, canopy/cover on motorized actuators etc as per check list.<br/>Responsibility: Head of EMD</p> <p>G) The locking mechanism of the Butterfly valve in position during disengagement of Actuator of Butterfly valve is as below:</p> <p>(i) The locking mechanism in the discharge butterfly valve using locking pin is feasible only in the following two cases:</p> <ol style="list-style-type: none"> <li>When the BFV is in fully opened condition</li> <li>When the BFV is in fully closed condition</li> </ol> <p>Only for the above two conditions the holes for locking pin in blade restraint mechanism are aligned thereby facilitating the insertion lock pin.<br/>The locking pin should be placed in position as per point no: A above.</p> <p>(ii) Given the interconnection of discharge ducts of CW pumps, the Actuator of BFV should be disengaged only after the upstream and downstream of the BFV is emptied. This is feasible only during Unit shutdown.</p> <p>(iii) For replacement/ maintenance of Hydraulic cylinder:</p> <ol style="list-style-type: none"> <li>Upstream/ downstream of BFV to be emptied.</li> <li>Supply of CW power pack to be isolated.</li> <li>Power pack oil to be drained, depressurized. Oil from hose connections at hydraulic cylinder to be drained.</li> </ol> |                  |                      |

|                                                                                                                                  |                  |                      |
|----------------------------------------------------------------------------------------------------------------------------------|------------------|----------------------|
| <b>NTPC</b>                                                                                                                      | <b>LMI</b>       | <b>SIMHADRI</b>      |
| <b>TITLE: Requirements for the Operation and Maintenance of Large Butterfly Valves in Cooling Water Systems at NTPC Simhadri</b> |                  |                      |
|                                                                                                                                  | <b>Rev: 6.00</b> | <b>Date:Dec'2023</b> |
| <b>REF NO. LMI/OGN/OPS/MECH/008/</b>                                                                                             |                  | <b>PAGE :4 OF 10</b> |

- d) The hydraulic cylinder can be removed from fixture when the valve is partially open. For this purpose, the dead weight of the BFV is to be supported with a robust pre-fabricated stand to prevent its downward movement. EOT crane in CW pump house to be used for positioning the dead weight on the pre-fabricated support stand.
- e) Hydraulic cylinder can be now removed from the fixture and new cylinder can be replaced. Before placing the new/ refurbished cylinder, it should have been hydraulically tested at manufacturing/repairing site for any oil leakages and for confirming stroke length as per OEM standards.
- f) Hydraulic oil hoses to be connected to the Cylinder. Oil top up to be done.
- g) System to be checked for any oil leakages.

Responsibility: Head of MM-OS, Head of MM-TMD

## **8.0 EXCEPTION REPORTING**

The exceptions ( free movement of pins, damage to valve blade/ body) encountered during the inspection are to be reported to Head of O&M as per the following format and to be taken up with corporate engg. & OEM

Responsibility: Head of MM-OS, Head of MM-TMD

| Description of exception | Valve Tag no | Location of exception | Remarks |
|--------------------------|--------------|-----------------------|---------|
|                          |              |                       |         |

## **9.0 RESPONSIBILITIES**

The responsibility of carrying out different tasks in relation to inspection and maintenance of integrity of large butterfly valve as detailed in LMI are with Head of MM-OFS, Head of MM-TMD

## **10.0 RECORD KEEPING**

The records to be maintained under this LMI include as follows.

| Sl.No | Record Description                                   | Maintained by | Frequency | Remarks |
|-------|------------------------------------------------------|---------------|-----------|---------|
| 1     | CW Discharge BFV                                     | MM            |           |         |
| 2     | CW Inter connection valve                            | MM            |           |         |
| 3     | Condenser I/L & O/L valve                            | MM            |           |         |
| 4     | CW line Valves / Riser Valves / Recirculation valves | MM            |           |         |

- a) Compliance record of the activities detailed under this LMI.
- b) Exception reporting.

The above records to be maintained by Head of MM-OS, Head of MM-TMD

**11.0 REVIEW: This** document shall be reviewed by HOP (Simhadri) on Two yearly basis.

|                                                                                                                                        |  |     |  |          |
|----------------------------------------------------------------------------------------------------------------------------------------|--|-----|--|----------|
| NTPC                                                                                                                                   |  | LMI |  | SIMHADRI |
| TITLE: Requirements for the Operation and Maintenance of Large Butterfly Valves in Cooling Water Systems at NTPC Simhadri    Rev: 6.00 |  |     |  |          |





|                                                                                                                                  |                  |                       |
|----------------------------------------------------------------------------------------------------------------------------------|------------------|-----------------------|
| <b>NTPC</b>                                                                                                                      | <b>LMI</b>       | <b>SIMHADRI</b>       |
| <b>TITLE: Requirements for the Operation and Maintenance of Large Butterfly Valves in Cooling Water Systems at NTPC Simhadri</b> |                  |                       |
|                                                                                                                                  | <b>Rev: 6.00</b> | <b>Date: Dec'2023</b> |
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### Annexure- 1

| Check                                                                                          | Responsibility | Frequency            | Remarks                                                   |
|------------------------------------------------------------------------------------------------|----------------|----------------------|-----------------------------------------------------------|
| Valve Seal Check                                                                               | MM & OPN       | Overhaul             |                                                           |
| Valve freeness                                                                                 | MM & OPN       | Overhaul             |                                                           |
| Valve Seal adjustment/Replacement                                                              | MM             | Overhaul             |                                                           |
| Valve Lever Arm & Counterweight healthiness                                                    | MM             | Overhaul             |                                                           |
| Valve Bearing Check/replacement if required                                                    | MM             | Overhaul/Opportunity |                                                           |
| Leaks from Valve drive & non drive shaft and replacement of seals if required                  | MM             | Overhaul/Opportunity |                                                           |
| Disc to shaft locking arrangement                                                              | MM             | Overhaul             |                                                           |
| Valve locking arrangement in open position/close position                                      | MM             | Overhaul             |                                                           |
| Cleaning of Oil tank and oil replacement if necessary                                          | MM             | Overhaul/Opportunity |                                                           |
| Check the quality of Hydraulic Oil & Record                                                    | MM/Chemistry   | 6months              | MI<100ppm;<br>Moisture<300ppm                             |
| Condition of Power Cylinder: Check for internal seal leaks, Chrome plating condition of piston | MM             | Overhaul/Opportunity | Replace power cylinder once in 8Y                         |
| Replace the Power Cylinder Seals                                                               | MM             | Overhaul/Opportunity | Seal Kits to be purchased from OEM/OES                    |
| Condition of Supply & Return hose connected to power cylinder                                  | MM             |                      | Replace hoses once in 5 years OEM/PARKER Make/ Gates Make |
| Ensure Healthiness of hydraulic pump and its accessories like drive coupling etc               | MM             | Overhaul/Opportunity |                                                           |
| Overhauling of Hydraulic Pump Motor                                                            | EMD            | Overhaul/Opportunity |                                                           |
| Hydraulic system P&I checks                                                                    | MM             | Overhaul/Opportunity |                                                           |
| Ensure healthiness of PRV, ball valves, drain valves etc.                                      | CnI            | Overhaul/Opportunity |                                                           |

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| NTPC                                                                                                                        | LMI                                           |                      | SIMHADRI                   |
|-----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|----------------------|----------------------------|
| TITLE: Requirements for the Operation and Maintenance of Large Butterfly Valves in Cooling Water Systems at NTPC Simhadri   |                                               |                      |                            |
| Rev: 6.00                                                                                                                   |                                               | Date:Dec’2023        |                            |
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| Ensure PRV setting as per OEM                                                                                               | CnI                                           | Overhaul/Opportunity |                            |
| Encsure healthiness of check valves & Flow control valves & solenoid valves                                                 | CnI                                           | Overhaul/Opportunity |                            |
| Check the condition of Hydraulic Unit Breather & Replace if necessary                                                       | MM                                            | Overhaul/Opportunity |                            |
| Replacement of suction and return Pressure line oil filters                                                                 | MM                                            | Overhaul/Opportunity |                            |
| Check for oil leaks in Hydraulic System & attend                                                                            | MM                                            | Overhaul/Opportunity |                            |
| Checking of Bladder N2 gas pressure, identify leaks if any & attend the same and fill to desired pressure & record          | MM-hoses, Pipes, Fittings<br>CnI -Instruments | Overhaul/Opportunity |                            |
| Monitoring of start frequency& running time /Frequency of bladder charging solenoids( Monitoring of bladder N2 GasPressure) | Opern & CnI                                   | 12W                  |                            |
| Replace bladders of accumulator, Identify leaks if any in bladder & attend same, fill to desired pressure & record.         | MM                                            | Overhaul/Opportunity | Replacement once in 5Years |
| Healthiness of Hand Pump                                                                                                    | MM                                            | Overhaul/Opportunity |                            |
| Healthiness of Valve Position & Feed back transmitters                                                                      | CnI                                           | Overhaul/Opportunity |                            |
| Healthiness of limit switches: Crack Open, Open, Creep, close, intermediate position, lock pin open, & lock pin close       | CnI                                           | Overhaul/Opportunity |                            |
| Check for Valve Open & close feed back                                                                                      | MM, CnI & Opern                               | Overhaul/Opportunity |                            |
| Note the timings of Valve, Emergency Closing, Normal closing                                                                | MM, CnI & Oeprn                               | Over haul            |                            |
| Ensure Hard Barricading around the counterweight operating zone.                                                            | M<br>M                                        | 4W check             |                            |

|                                                                                                                                             |            |                      |
|---------------------------------------------------------------------------------------------------------------------------------------------|------------|----------------------|
| <b>NTPC</b>                                                                                                                                 | <b>LMI</b> | <b>SIMHADRI</b>      |
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**Annexure – II: Checklist for Motorized Butterfly Valve**

| <b>Check</b>                                                                                           | <b>Responsibility</b> | <b>Frequency</b>                      | <b>Remarks</b> |
|--------------------------------------------------------------------------------------------------------|-----------------------|---------------------------------------|----------------|
| Valve Seal Check                                                                                       | MM & OPN              | Overhaul                              |                |
| Valve freeness                                                                                         | MM & OPN              | Overhaul                              |                |
| Valve Seal adjustment/Replacement                                                                      | MM                    | Overhaul                              |                |
| Valve actuator connecting bush & fasteners                                                             | MM                    | Overhaul                              |                |
| Valve Bearing Check/replacement if required                                                            | MM                    | Overhaul                              |                |
| Leaks from Valve drive & non drive shaft and replacement of seals if required                          | MM                    | Overhaul                              |                |
| Disc and shaft locking arrangement                                                                     | MM                    | Overhaul                              |                |
| Gear box Servicing                                                                                     | MM                    | Overhaul                              |                |
| Actuator servicing                                                                                     | EMD                   | Overhaul                              |                |
| Setting of Open/close torque switches                                                                  | EMD & OPN             | Overhaul/Opportunity                  |                |
| Healthiness of Control & 3Ph. Power supply (MCC Disturbance) and local/remote selector switch at local | EMD                   | Overhaul/Opportunity                  |                |
| Healthiness of lever engagement for manual operation of Actuator Valve                                 | MM, EMD & OPN         | Overhaul/Opportunity                  |                |
| Canopy/Cover on motorized actuators to prevent water ingress as per site conditions                    | EMD & OPN             | 4W check                              |                |
| Valve Pit Healthiness to be ensured                                                                    | OPN                   | 4W check/ Weekly check during monsoon |                |

|                                                                                                                                  |                  |                       |
|----------------------------------------------------------------------------------------------------------------------------------|------------------|-----------------------|
| <b>NTPC</b>                                                                                                                      | <b>LMI</b>       | <b>SIMHADRI</b>       |
| <b>TITLE: Requirements for the Operation and Maintenance of Large Butterfly Valves in Cooling Water Systems at NTPC Simhadri</b> |                  |                       |
|                                                                                                                                  | <b>Rev: 6.00</b> | <b>Date: Dec'2023</b> |
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**Annexure – III: Checklist for Manual Butterfly Valve**

| <b>Check</b>                                                                  | <b>Responsibility</b> | <b>Frequency</b>     | <b>Remarks</b> |
|-------------------------------------------------------------------------------|-----------------------|----------------------|----------------|
| Valve Seal Check                                                              | MM & OPN              | Overhaul/Opportunity |                |
| Valve freeness                                                                | MM & OPN              | Overhaul/Opportunity |                |
| Valve Seal adjustment/Replacement                                             | MM                    | Overhaul/Opportunity |                |
| Valve over travel locking mechanism                                           | MM                    | Overhaul/Opportunity |                |
| Valve Bearing Check/replacement if required                                   | MM                    | Overhaul/Opportunity |                |
| Leaks from Valve drive & non drive shaft and replacement of seals if required | MM                    | Overhaul/Opportunity |                |
| Disc and shaft locking arrangement                                            | MM                    | Overhaul/Opportunity |                |
| Gear box Servicing                                                            | MM                    | Overhaul/Opportunity |                |

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